

AI Gym & Fitness Assistant

Internship Project Report

Developed as part of the Industry Internship at:
Unlox Academy and Trivion Technology

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Role: Project Intern

1. Project Overview

The AI Gym & Fitness Assistant is a unified, AI-powered fitness ecosystem developed during the internship at Unlox Academy and Trivion Technology. Rather than treating workout tracking, diet planning, motivation, and equipment monitoring as separate tools, this project brings all of them into a single platform, acting as a smart personal trainer, dietician, motivator, and data-driven fitness manager for the user.

The system is built around seven core AI modules covering computer-vision-based workout detection, LLM-driven diet coaching, simulated IoT gym equipment monitoring, behavioral risk prediction, a conversational fitness companion, automated performance scoring, and a recommendation engine — all served through a FastAPI backend and accessible via both a deployed Streamlit dashboard and a React/Next.js web frontend.

2. Objectives

- Design and implement a modular AI ecosystem covering the full fitness journey: detection, diet, behavior, IoT-simulated hardware, and conversational support.
- Demonstrate real, working integration between a computer-vision module and a backend analytics module, without manual data re-entry.
- Deploy the system publicly (not just localhost) so it can be demonstrated and used from any device.
- Provide two parallel frontend experiences (Streamlit for rapid delivery, React/Next.js for a production-style UI) served by the same backend.
- Document, test, and verify each module independently and as an integrated whole.

3. System Architecture

The architecture follows a layered design: a user-facing frontend layer (Streamlit + React/Next.js), a FastAPI backend layer hosted on Render, seven AI modules, external services (Google Gemini API, MediaPipe pose model), and a deployment layer (Render, Streamlit Community Cloud, and an available Docker Compose setup).

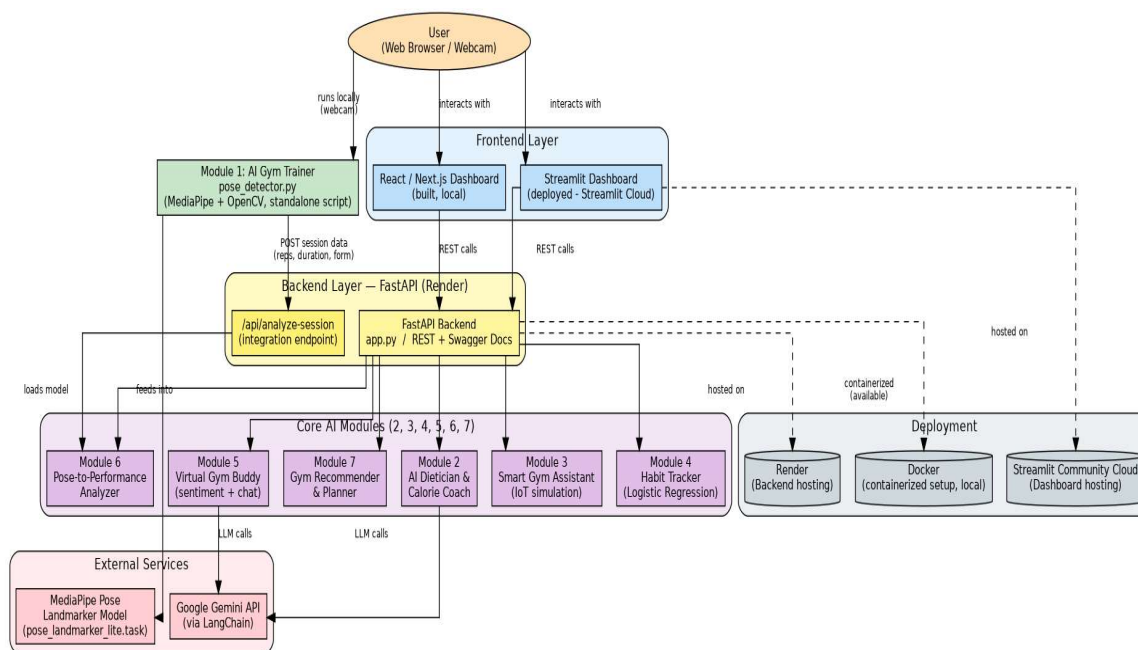


Figure 1: System architecture — data flow from user through frontend, backend, AI modules, and external services.

The one live, automatic integration in the system is between Module 1 (AI Gym Trainer) and Module 6 (Pose-to-Performance Analyzer): when a webcam session ends, the session's rep count, duration, and form feedback are POSTed directly to the backend's /api/analyze-session endpoint, which immediately returns a performance score — with no manual data entry required. The remaining modules (2, 3, 4, 5, 7) operate as independent tools within the same dashboard and backend, in line with the project's modular design.

4. Technology Stack

Layer	Technology Used
Frontend	Streamlit (deployed); React.js / Next.js (built, local)
Backend	Python, FastAPI
AI / ML	Google Gemini API (via LangChain), MediaPipe Pose Landmarker, OpenCV, scikit-learn (Logistic Regression)
Deployment	Render (backend), Streamlit Community Cloud (dashboard), Docker (available)
Version Control	Git / GitHub

Note: the original proposal considered a broader stack (TensorFlow/PyTorch, MongoDB/PostgreSQL, MQTT/Node-RED, Hugging Face/OpenAI, AWS S3/Firebase, Plotly/D3.js). The final implementation converged on a leaner, faster-to-ship stack — MediaPipe + scikit-learn for the AI/ML side and Gemini for the conversational and generative pieces — which was sufficient to meet every module's functional goal within the internship timeline.

5. Core AI Modules

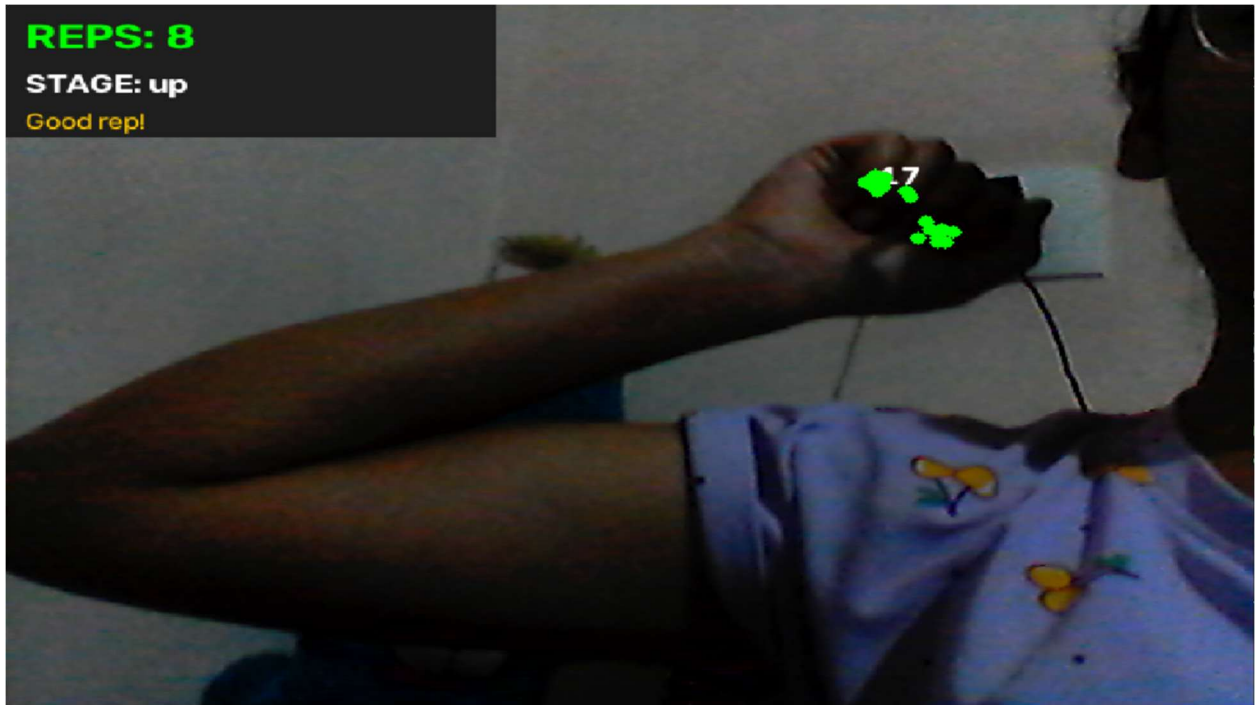
5.1 Module 1 — AI Gym Trainer (Workout Detection & Feedback)

Runs as a standalone Python script (ai_modules/pose_detector.py) using MediaPipe's Pose Landmarker and OpenCV to track the user's arm in real time through a webcam. It counts bicep curl repetitions, monitors elbow angle to detect poor form (elbow drift), and gives live audio and visual feedback — a confirmation beep for a good rep and a warning beep on form drift. It runs as a native script rather than a browser page because real-time webcam frame processing is better suited to a local script than a browser environment; it integrates with the rest of the system purely through the backend API.

REPS: 8

STAGE: up

Good repl



REPS: 23

STAGE: down

Good repl





5.2 Module 2 — AI Dietician & Calorie Coach

Takes the user's weight, height, goal, and dietary preferences, calculates BMI and a daily calorie target, and uses the Gemini API to generate a personalized meal plan and grocery list. A verified sample run for an underweight profile (BMI 17.3, muscle-gain goal) produced:

- Estimated daily calorie target: ~2,600–2,800 kcal
- A full one-day, four-meal plan (breakfast, lunch, snack, dinner) built around nutrient-dense, high-calorie, high-protein foods
- A grocery list grouped into Produce, Protein & Dairy, and Pantry & Grains

The response also included an automatic disclaimer recommending the user consult a physician or registered dietitian before making major dietary changes.

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AI Gym & Fitness Assistant — Dashboard

Diet Coach

Behavior Risk

Gym Buddy

Smart Gym

Recommendations

Session Performance

AI Dietician & Calorie Coach

Weight (kg)

50

Press Enter to apply

-

+

Height (cm)

170.00

-

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Goal

weight loss

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Dietary preferences (e.g., vegetarian, vegan, no restrictions)

no restrictions

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Weight (kg)

50.00

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Height (cm)

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Goal

weight loss

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Dietary preferences (e.g., vegetarian, vegan, no restrictions)

no restrictions

Generate Diet Plan

BMI

17.3

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Hello! As your AI Dietician and Calorie Coach, I am here to help you navigate your nutritional journey safely and effectively.

1. Assessment & Important Note

Your current BMI is 17.3, which falls into the underweight category. While I understand you have expressed a goal of weight loss, focusing strictly on losing weight when you are already underweight can pose risks to your immune system, bone density, and overall health. Instead, I strongly encourage a focus on gentle body recomposition, nourishing your body, and building strength.

Disclaimer: I am an AI, not a doctor. Please consult a healthcare professional or a registered dietitian before making significant changes to your diet, especially when managing body weight from an underweight status.

2. Estimated Daily Calorie Target ↔

To support safe body nourishment and healthy weight stabilization, a gentle maintenance or slight caloric surplus is typically recommended for someone with your current BMI. However, to respect your current goals while prioritizing safety, a balanced target of approximately **2,000 to 2,200 calories per day** is a healthy baseline to discuss with a professional.

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3. One-Day Sample Meal Plan

This balanced, nutrient-dense menu provides sustained energy and healthy fats:

- **Breakfast:** Oatmeal made with whole milk, topped with 1 sliced banana, 1 tablespoon of peanut butter, and a sprinkle of chia seeds.
- **Lunch:** Mediterranean wrap: A large whole-wheat tortilla filled with sliced grilled chicken breast, hummus, feta cheese, cucumber, mixed greens, and a drizzle of olive oil.
- **Snack:** A handful of mixed nuts (almonds and walnuts) paired with an apple or pear.
- **Dinner:** Baked salmon fillet served alongside a generous portion of quinoa and roasted broccoli drizzled with olive oil and garlic.

4. Short Grocery List

- Whole milk (or preferred milk alternative)
- Bananas, apples/pears
- Chia seeds

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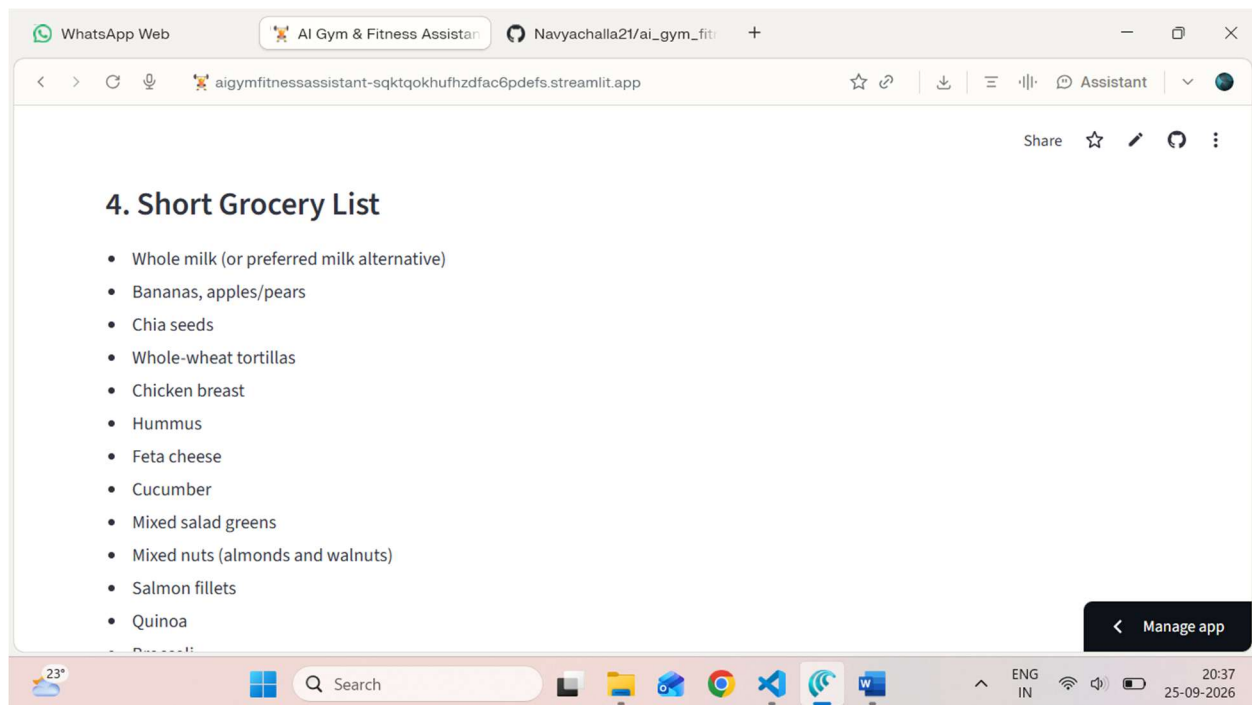
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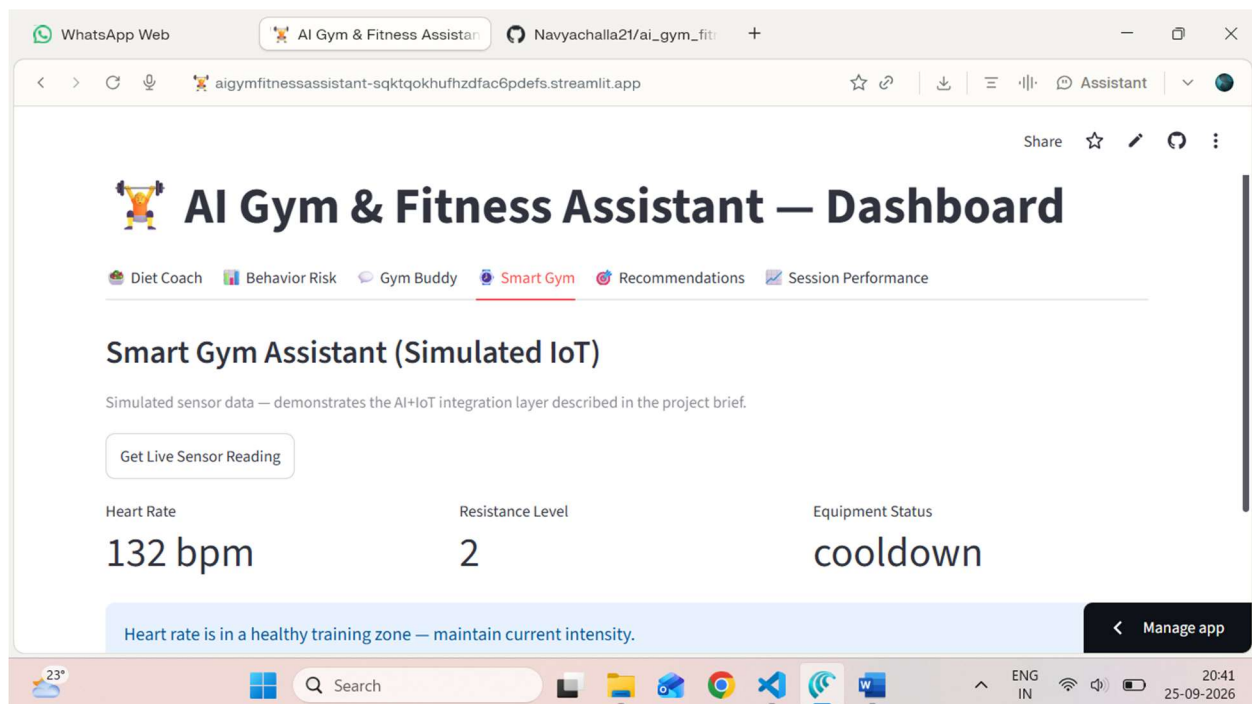
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5.3 Module 3 — Smart Gym Assistant (AI + IoT Simulation)

Simulates IoT-enabled gym equipment sensor data — heart rate, resistance level, and equipment status — and gives real-time intensity or rest advice. In a verified live reading, the module reported a heart rate of 121 bpm, resistance level 5, and equipment status "active", correctly classifying 121 bpm as within a healthy training zone and advising the user to maintain current intensity.



5.4 Module 4 — AI Fitness Habit Tracker (Behavioral AI)

Uses a trained logistic regression model to predict the risk that a user will skip their next workout, based on recent engagement patterns (days since last workout, average sessions per week, average session completion percentage), and returns a motivational nudge alongside the prediction. A verified test case — 2 days since last workout, 4 average sessions per week, 85% average completion — was correctly classified as Low skip risk (1%).

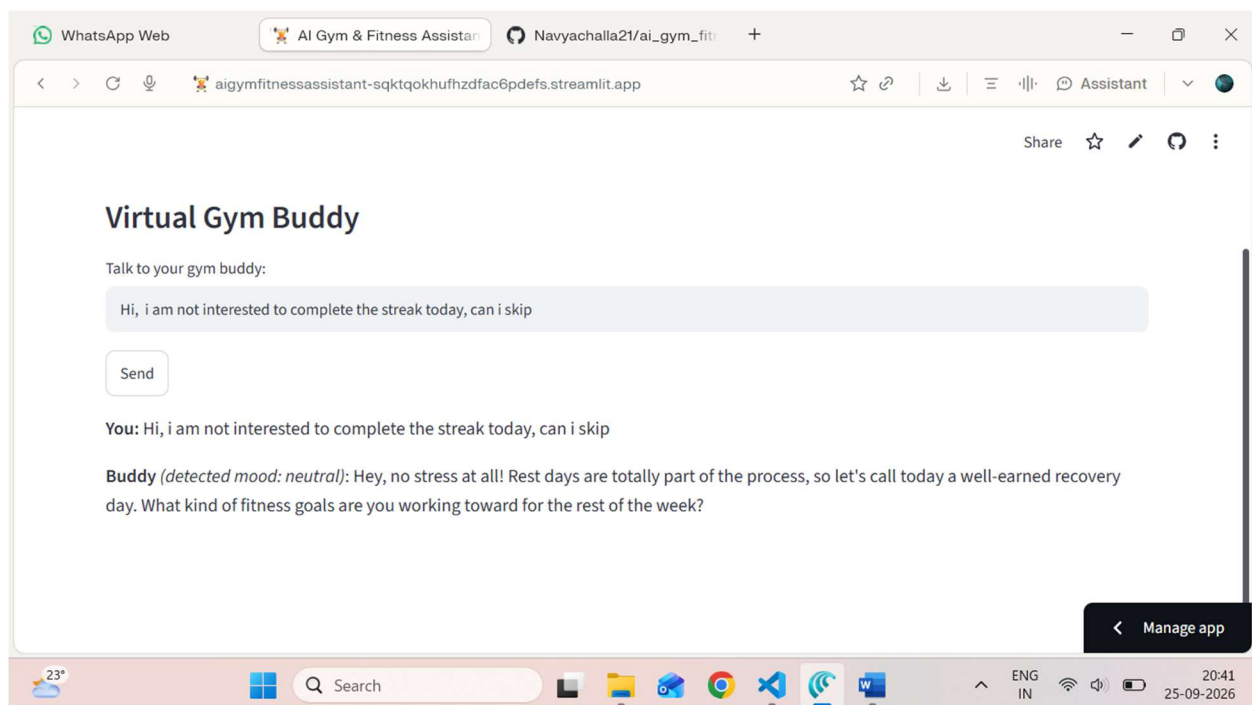
The dashboard, titled "AI Gym & Fitness Assistant — Dashboard", features a navigation bar with tabs: Diet Coach, Behavior Risk (selected), Gym Buddy, Smart Gym, Recommendations, and Session Performance. The "Behavior Risk" section includes three sliders for inputting user data:

- Days since last workout: 5
- Average sessions per week: 4.00
- Average session completion %: 85

A "Manage app" button is located at the bottom right of the input section. The bottom screenshot shows the "Predict Skip Risk" button, which triggers a prediction of "High" skip risk, accompanied by a green upward arrow and "100%". A motivational message states: "We miss you! A quick 15-minute session today keeps your streak alive 🍌".

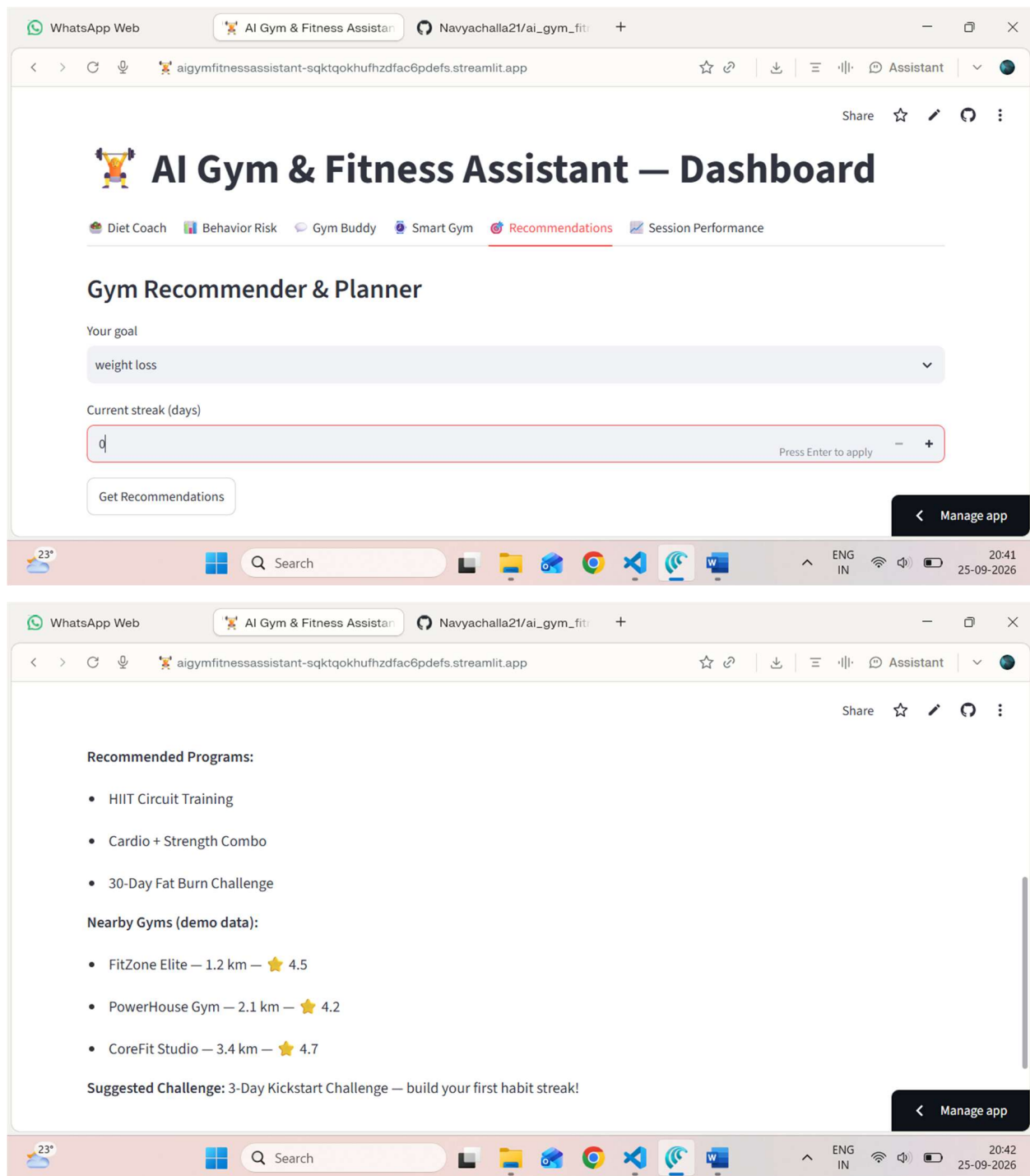
5.5 Module 5 — Virtual Gym Buddy (AI Chat Companion)

A Gemini-powered conversational companion with keyword-based sentiment detection, designed to respond supportively based on the user's detected mood. A verified multi-turn conversation showed the buddy greeting the user warmly, asking about fitness goals, and maintaining a consistent, encouraging persona ("mood: neutral") across turns, including handling an out-of-scope question (asking the buddy's own name) gracefully by redirecting back to the user's goals.



5.6 Module 6 — Pose-to-Performance Analyzer

Scores a completed workout session into a Performance Score, Form Quality percentage, and a qualitative Rating, based on reps completed, session duration, and the ratio of positive to total feedback events. A verified sample session — 12 reps in 120 seconds, with 10 of 12 feedback events being "Good rep!" — produced a Performance Score of 66, Form Quality of 83.3%, and a Rating of "Good". In production, this module receives its input automatically from Module 1 rather than manual entry.



6. Deployment

The backend is deployed on Render as a FastAPI service with interactive Swagger API documentation. The Streamlit dashboard is deployed on Streamlit Community Cloud and communicates with the live Render backend (not localhost) — verified end-to-end, including real Gemini AI responses served from a public URL. The React/Next.js frontend was built and verified locally, producing results consistent with the Streamlit dashboard for

all six non-webcam modules. A Docker Compose configuration is included in the repository for containerized deployment of both backend and frontend, and is planned for full use post-submission.

Because the Render backend runs on a free tier that spins down after periods of inactivity, the first request after idle time can take roughly 30–50 seconds while the server restarts — a known, accepted trade-off of the free hosting tier rather than an application defect.

7. Testing Summary

- Diet Coach: verified correct BMI calculation, category classification, and AI-generated meal plans across multiple goals (muscle gain, weight loss).
- Behavior Risk: verified skip-risk predictions and nudges across low/medium/high risk scenarios.
- Gym Buddy Chat: verified sentiment detection (positive/neutral/negative) and contextual AI replies across a multi-turn conversation.
- Smart Gym: verified simulated sensor readings and matching advice (e.g., high heart rate → rest recommendation).
- Recommendations: verified goal-based program suggestions, demo gym listings, and challenge suggestions.
- Session Performance: verified performance score, form quality %, and rating calculations from sample session data.
- AI Gym Trainer (Module 1): tested live with a webcam and physical dumbbells — verified real-time rep counting, elbow angle tracking, form-drift warning (audio beep), and good-rep confirmation (audio beep).
- Module 1 → Module 6 integration: verified that ending a live webcam session automatically posts session data to the backend and returns a live performance score, with no manual re-entry.
- Deployment: verified the live Streamlit Cloud dashboard communicates end-to-end with the live Render backend from a public URL.
- Cross-platform parity: verified all six non-webcam modules produce consistent results across the Streamlit and React/Next.js frontends.

8. Known Limitations & Challenges

- Module 1 runs as a standalone Python script rather than a browser page, since real-time webcam frame processing with MediaPipe/OpenCV is better suited to a native script than a browser environment; it still integrates with the rest of the system through the backend API.
- Render's free-tier backend spins down after roughly 15 minutes of inactivity, so the first request after idle time takes about 30–50 seconds while the server restarts.
- Docker containerization is complete and available in the repository, but was not used as the primary deployment method within the project timeline; the Streamlit Cloud + Render combination was used instead for the live demo.

9. Conclusion

The AI Gym & Fitness Assistant successfully delivers on its core objective: a single platform that unifies workout detection, diet planning, behavioral prediction, simulated IoT monitoring, conversational motivation, and performance analysis. All seven modules were implemented, deployed, and independently verified, and the one designed integration point — live webcam sessions feeding directly into automated performance scoring — was demonstrated working end-to-end. The result is a functioning, publicly accessible prototype of an all-in-one AI fitness ecosystem, built within the scope of the Unlox Academy internship.

10. Repository & Live Links

- Live Dashboard (Streamlit): <https://aigymfitnessassistant-sqktqokhufhzdfac6pdefs.streamlit.app>
- Backend API (Render): <https://ai-gym-backend-bwiz.onrender.com>
- API Docs (Swagger): <https://ai-gym-backend-bwiz.onrender.com/docs>
- Source Repository: https://github.com/Navyachalla21/ai_gym_fitness_assistant